📘 Propagation Channel Scenarios (5G)

✅ 1. Definition

- Propagation channel scenarios refer to **different real-world environments in which wireless signals travel from transmitter to receiver**.

🎯 Simple Meaning

👉 Signal behaves differently depending on location:

- City 🏢
- Village 🌾
- Inside building 🏠
- Moving vehicle 🚗

📊 Basic Diagram

📡 →→→ Environment →→→ 📱
(Urban / Rural / Indoor / Mobility)

🔷 2. Types of Propagation Scenarios

🏢 1. Urban Scenario (City Area)

📌 Description

- Area with **many buildings, towers, obstacles**

📌 Characteristics

- Strong **reflection**
- Severe **multipath propagation**
- High **interference**
- Signal blockage

📊 Diagram

📡 → 🏢🏢 → 📱
Reflection & blockage

🧠 Example

- Signal in Hyderabad city between tall buildings

🌾 2. Rural Scenario (Village Area)

📌 Description

- **Open areas**, fewer obstacles

📌 Characteristics

- Long-distance communication
- Less interference
- Mostly **Line-of-Sight (LOS)**

📊 Diagram

📡 →→→→→→→→→ 📱
(Open field)

🧠 Example

- Signal in villages or highways

🏠 3. Indoor Scenario

📌 Description

- Communication **inside buildings**

📌 Characteristics

- Signal blocked by walls
- High **attenuation**
- Reflection dominant

📊 Diagram

📡 → 🏠 → 📱
Signal weak inside

🧠 Example

- Weak signal inside room

🚗 4. High Mobility Scenario

📌 Description

- User is moving fast (car/train)

📌 Characteristics

- **Doppler effect**
- Rapid signal changes
- Frequent handover

📊 Diagram

📡 →→→ 🚗 → 📱
Moving user

🧠 Example

- Using phone in train

🏙️ 5. Dense Urban / Small Cell Scenario (Advanced)

📌 Description

- Highly crowded areas

📌 Characteristics

- Very high user density
- Small cell deployment

- High data demand

🧠 Example

- Stadium, malls

🔷 3. Comparison of Scenarios

Scenario	Obstacles	Range	Interference
Urban	High	Short	High
Rural	Low	Long	Low
Indoor	Medium	Short	Medium
Mobility	Variable	Medium	High

🔷 4. Importance of Studying Scenarios

- Helps design better networks
- Improves signal quality
- Helps in planning base stations
- Reduces interference